LAB 2

#1 Create a structure in C++ containing the details of students as details below and a main function to execute the structure…

```
#include <iostream>
#include <string>
using namespace std;

    struct
    student{
    string name;
    int roll;
    string degree;
    char hostel;
    float cg;

        void adddetails()
        {
        cout<<"Enter
name,rollno,degree,hostel,cgpa: ";
                cin>>name>>roll>>degree>>hos
        tel>>cg;
        }
        void updatedetails()
        {
                adddetails();
        }
        void updatecgpa()
        {
                cout<<"Enter new cgpa: ";
                cin>>cg;
        }
            void updatehostel(){
        cout<<"Enter new hostel: ";
        cin>>hostel;
        }
            void displaydetails(){
        cout<<"\nName: "<<name;
        cout<<"\nRollNo: "<<roll;
```

```
PS C:\Users\lenovo\Desktop\OOPs> cd "c:\Users\
Enter no of students: 2
Enter details of student 1
Enter name,rollno,degree,hostel,cgpa: Gauri 1
Enter details of student 2
Enter name,rollno,degree,hostel,cgpa: Gauri 2

Student 1Details:

Name: Gauri
RollNo: 1
Degree: Civil
Hostel: A
Current CGPA: 9

Student 2Details:

Name: Gauri
RollNo: 2
Degree: Mech
Hostel: B
Current CGPA: 6.7
PS C:\Users\lenovo\Desktop\OOPs\a1>
```

```
                cout<<"\nCurrent CGPA:
        "<<cg<<endl;
        }
};

int main(){
        int n;
        cout<<"Enter no of students: ";
    cin>>n; student s[30];
        for(int i=0;i<n;i++){
        cout<<"Enter details of student
"<<i+1<<endl;
                s[i].adddetails();
        }
        for(int j=0;j<n;j++){
                cout<<"\nStudent
        "<<j+1<<"Details:
"<<endl;
                s[j].displaydetails();
        }
        return 0;
}
```

#2 Differentiate between public and private access/scope. Perform Q1 with class instead of structure with having the data members private and some member functions in private scope and some in public scope.

Public scope variables/functions can be accessed from anywhere in the program, by anyone. Whereas those that are private can be accessed within its specific class only.
Eg. Your bank details will be private but the functions of making payments, transfer etc are accessible to the public in a banking website.

```
#include <iostream>
#include <string>
using namespace std;

    class student{
    private:
        string
    name; int
    roll; string
    degree; char
    hostel; float
    cg;

        public:
        void adddetails()
        {
        cout<<"Enter
name,rollno,degree,hostel,cgpa: ";

cin>>name>>roll>>degree>>hostel>>cg;
        }
        void updatedetails()
        {
                adddetails();
        }
        void updatecgpa()
        {
                cout<<"Enter new cgpa: ";
                cin>>cg;
        }
            void updatehostel(){
        cout<<"Enter new hostel: ";
        cin>>hostel;
        }
            void displaydetails(){
```

```
Enter no of students: 2
Enter details of student 1
Enter name,rollno,degree,hostel,cgpa: ga
Enter details of student 2
Enter name,rollno,degree,hostel,cgpa: ma

Student 1Details:

Name: gauri
RollNo: 241
Degree: civil
Hostel: q
Current CGPA: 9.8

Student 2Details:

Name: mansha
RollNo: 234
Degree: enc
Hostel: q
Current CGPA: 9.6
```

```
        }
};

int main(){
        int n;
        cout<<"Enter no of students: ";
    cin>>n; student s[30];
        for(int i=0;i<n;i++){
        cout<<"Enter details of student
"<<i+1<<endl;
                s[i].adddetails();
        }
        for(int j=0;j<n;j++){
                cout<<"\nStudent
        "<<j+1<<"Details:
"<<endl;
                s[j].displaydetails();
        }
        return 0;
}
```

#3 Create a code snippet that illustrates the following: calling of private member functions inside public member function

```
#include <iostream>
using namespace std;
class xyz
{
        private:
        void SecretFn()
        {
        cout<<"welcome to the secret
function!";
        };
        public:
        void GetSecret()
        {
                SecretFn();
        }
};
```

```
welcome to the secret function!
PS C:\Users\lenovo\Desktop\DSA>
```

```
int main()
{
        xyz t;
    t.GetSecret();
}
```

#4 Define a class Rectangle with variables width and height of integer type along with following:

(a) void getdata() to initialize object values

(b) Void calculatearea() to calculate and display the area

```
#include <iostream>
using namespace std;

class Rectangle
{
        private:
        int width, height;

        public:
        void getData()
        {
                cout<<"Enter width and
        height: ";
                cin>>width>>height;
        }
        void calculateArea()
        {
        cout<<"Area =
"<<width*height<<endl;
        }
};
int main()
{
        Rectangle r1;
    r1.getData();
    r1.calculateArea();
}
```

```
Enter width and height: 2 4
Area = 8
PS C:\Users\lenovo\Desktop\DSA>
```

#5 Define a class Complex with variables real and imaginary along with following:
(a) void setComplex (float, float) to initialize object values.
(b) void displayComplex() to show the complex number
Pass and return objects to calculate sum of two complex numbers. Display the sum.

```
#include <iostream>
using namespace std;

class Complex {
private:
        float real;
    float imaginary;

public:
        void setComplex(float r, float i) {
                real =
        r; imaginary =
        i;
        }
            void
        displayComplex() { if
        (imaginary >= 0) {
            cout << real << " + " <<
imaginary << "i" << endl;
                } else {
                        cout << real << " -
            " <<
-imaginary << "i" << endl;
                }
        }

            Complex sum(Complex c1, Complex
        c2) { Complex temp;
                temp.real = c1.real +
        c2.real; temp.imaginary =
        c1.imaginary +
c2.imaginary;
                return temp;
        }
```

```
Enter real and imaginary parts of first com
1 2
Enter real and imaginary parts of second co
3 4
First complex number: 1 + 2i
Second complex number: 3 + 4i
Sum of the two complex numbers: 4 + 6i
```

```
        float r1, i1, r2, i2;

    cout << "Enter real and imaginary parts
of first complex number: " << endl;
        cin
    >>r1;
    cin>>i1;
        c1.setComplex(r1, i1);
   cout << "Enter real and imaginary parts
of second complex number: " << endl;
        cin
    >>r2;
    cin>>i2;
        c2.setComplex(r2, i2);
        cout << "First complex number: ";
        c1.displayComplex();
        cout << "Second complex number: ";
        c2.displayComplex
    (); c3 = c3.sum(c1,
    c2);
    cout << "Sum of the two complex
numbers: ";
        c3.displayComplex
    (); return 0;
}
```